

ID–Cleft Continuum Derivation

Mathematical preliminaries and continuum assumptions

This section collects the constitutive laws and the structural assumptions that are used when passing from the discrete, patch-based conservation laws to their continuum (field) form. They are stated here once so that the derivation below can be read without interruption; the derivation itself preserves the original variables and notation.

Geometry and notation

The intercalated-disc (ID) membrane is modelled as a two-dimensional surface Ω embedded in $\mathbb{R}^3$. It is partitioned into M disjoint patches $\{w_j\}_{j=1}^M$; patch w_j has area a_j and a one-dimensional boundary curve ∂w_j with arclength element $d\ell$ and in-surface outward unit normal $\mathbf{n}$. Surface differential operators on Ω are written ∇_Ω (surface gradient) and $\nabla_\Omega \cdot$ (surface divergence). Capital symbols ($C, I, g/M, \dots$) denote per-patch (lumped) quantities, and the corresponding lower-case / barred symbols ($c, i, \bar{g}, \dots$) denote the associated surface densities obtained in the continuum limit. The index i labels the junction (cleft), j the patch within a junction, k a neighbouring patch, and s the ionic species. Subscripts pre, post, m, c, b refer respectively to the pre-junctional membrane, the post-junctional membrane, the myoplasm, the cleft, and the bulk reservoir. Here R is the gas constant, T the absolute temperature, F Faraday’s constant, z_s the valence of species s , and s_* a unit reference concentration.

Constitutive laws

Law 1 (Ohmic electrical conduction). The electric (charge) flux density is proportional to the gradient of the relevant potential,

$$\mathbf{J} = -\sigma \nabla \phi,$$

with σ the (surface) electrical conductivity. For the cleft this is generalised to an electro-diffusive flux driven by the electrochemical potential, $\mathbf{J}_c = -\sigma_c \nabla \eta_c^s$.

Law 2 (Surface divergence theorem). For a flux density $\mathbf{J}$ tangent to Ω and a patch w_j ,

$$\int_{\partial w_j} \mathbf{J} \cdot \mathbf{n} d\ell = \int_{w_j} \nabla_\Omega \cdot \mathbf{J} dA.$$

This converts the net current leaving a patch through its boundary into a surface divergence over the patch.

Law 3 (Electrochemical potential / Nernst–Planck driving force). The electrochemical potential of species s in the cleft and in the bulk are

$$\eta_c^s = \phi_c + \frac{RT}{z_s F} \ln \frac{s_c^{(s)}}{s_*}, \quad \eta_b^s = \phi_b + \frac{RT}{z_s F} \ln \frac{s_b^{(s)}}{s_*},$$

so that an inter-compartment current is driven by the difference of electrochemical potentials, $i = g(\eta_{\text{from}}^s - \eta_{\text{to}}^s)$, i.e. by the sum of an electrical ($\nabla\phi$) and a diffusive ($\nabla \ln s$) contribution.

Law 4 (Faraday charge–mass equivalence). The molar rate of change of a species equals the net ionic current carrying that species divided by $z_s F$ (per unit cleft volume). This is the relation used to turn the cleft charge balance into an evolution equation for the ionic concentration $s_c^{(s)}$.

Constitutive closure (kept symbolic). The cleft electro-diffusive conductivity $\sigma_c^{(s)}$ is treated throughout as a given coefficient and is *not* expanded into any explicit closure (e.g. a Nernst–Einstein form); it is retained as a symbol so that the field equations stay in their original variables.

Continuum assumptions

Assumption 1 (Homogenisation / mesh refinement). As $M \rightarrow \infty$ and $|w_j| \rightarrow 0$, the lumped patch quantities converge to smooth surface fields on Ω , and a per-patch balance that holds for the integral over every w_j holds pointwise on Ω .

Assumption 2 (Local ID coupling). The inter-patch (ID and cleft) conductances $g_{\text{ID},i,j,k}$, $g_{c,i,j,k}$ couple neighbouring patches only, and scale with patch size so that the nearest-neighbour sums converge to a second-order surface operator, i.e. to $\nabla_\Omega \cdot (\sigma \nabla_\Omega \cdot)$. Long-range (non-local) coupling is excluded.

Assumption 3 (Charge-conservation constraint / quasi-electroneutrality). At each point of Ω the cleft satisfies an instantaneous charge balance: the capacitive and ionic currents delivered by the two facing membranes are balanced by the bulk and lateral cleft currents. The cleft-potential equation plays the role of this constraint, while the per-species concentrations evolve in time.

Assumption 4 (Bulk reservoir). The extracellular bulk is a reservoir of fixed concentration $s_b^{(s)}$ and fixed potential, taken as the reference $\phi_b = 0$ unless stated otherwise.

Assumption 5 (Separation of capacitive and ionic transport). Capacitive (displacement) currents charge the membranes but do not transport ions. Hence they appear in the charge (potential) balances but are excluded from the per-species mass-conservation equation for $s_c^{(s)}$.

Pre-junctional field equation

$$\begin{aligned} \frac{g_{\text{myo}}}{M} (\phi_{\text{pre},i,j} - \phi_{m,i}) + C_{\text{pre},i,j} \frac{dV_{\text{pre},i,j}}{dt} + I_{\text{pre},i,j} + g_{\text{gap},i,j} \\ (\phi_{\text{pre},i,j} - \phi_{\text{post},i+1,j}) + \sum_{\substack{k=1 \\ k \neq j}}^M g_{\text{ID},i,j,k} (\phi_{\text{pre},i,j} - \phi_{\text{pre},i,k}) = 0. \end{aligned} \quad (\text{S1d})$$

This represents the conservation law in patch j , junction i . We set each patch w_j to have area a_j . Equation (S1d) can be written as

$$\int_{w_j} \bar{g}_{\text{myo}} (\phi_{\text{pre}} - \phi_m) + c_{\text{pre}} \frac{dV_{\text{pre}}}{dt} + i_{\text{pre}} + \bar{g}_{\text{gap}} (\phi_{\text{pre}} - \phi_{\text{post}}) dA + \sum_{\substack{k=1 \\ k \neq j}}^M g_{\text{ID},i,j,k} (\phi_{\text{pre},i,j} - \phi_{\text{pre},i,k}) = 0.$$

Here,

$$I_{\text{pre,out},j} = \sum_{\substack{k=1 \\ k \neq j}}^M g_{\text{ID},i,j,k} (\phi_{\text{pre},i,j} - \phi_{\text{pre},i,k}).$$

This represents the overall current going out of w_j . If we set the current flux density of w_j as $\mathbf{J}$, by definition

$$I_{\text{pre,out},j} = \int_{\partial w_j} \mathbf{J} \cdot \mathbf{n} d\ell.$$

Based on Ohm's law

$$\mathbf{J} = -\sigma_{\text{pre}} \nabla \phi_{\text{pre}}.$$

We can write

$$I_{\text{pre,out},j} = - \int_{\partial w_j} \sigma_{\text{pre}} \nabla \phi_{\text{pre}} \cdot \mathbf{n} d\ell = - \int_{w_j} \nabla_{\Omega} \cdot (\sigma_{\text{pre}} \nabla \phi_{\text{pre}}) dA.$$

Hence, we can write (S1d) as

$$\int_{w_j} \bar{g}_{\text{myo}}(\phi_{\text{pre}} - \phi_m) + c_{\text{pre}} \frac{dV_{\text{pre}}}{dt} + i_{\text{pre}} + \bar{g}_{\text{gap}}(\phi_{\text{pre}} - \phi_{\text{post}}) - \nabla_{\Omega} \cdot (\sigma_{\text{pre}} \nabla \phi_{\text{pre}}) dA = 0.$$

When $M \rightarrow \infty$, $|w_j| \rightarrow 0$,

$$\bar{g}_{\text{myo}}(\phi_{\text{pre}} - \phi_m) + c_{\text{pre}} \frac{dV_{\text{pre}}}{dt} + i_{\text{pre}} + \bar{g}_{\text{gap}}(\phi_{\text{pre}} - \phi_{\text{post}}) - \nabla_{\Omega} \cdot (\sigma_{\text{pre}} \nabla \phi_{\text{pre}}) = 0.$$

Post-junctional field equation

By applying the same method

$$\bar{g}_{\text{myo}}(\phi_{\text{post}} - \phi_m) + c_{\text{post}} \frac{dV_{\text{post}}}{dt} + i_{\text{post}} + \bar{g}_{\text{gap}}(\phi_{\text{post}} - \phi_{\text{pre}}) - \nabla_{\Omega} \cdot (\sigma_{\text{post}} \nabla \phi_{\text{post}}) = 0.$$

Axial membrane field equation

$$c_m \partial_t \phi_m + I_m + \sum_{N_{\text{junctions}}} [g_{\text{myo}}(\phi_m - \langle \phi_{\text{pre}} \rangle) + g_{\text{myo}}(\phi_m - \langle \phi_{\text{post}} \rangle)] = 0.$$

Cleft potential and ionic concentration equations

The original cleft potential and ionic concentration ODE can be written as

$$C_{r,i,j} \frac{dV_{r,i,j}}{dt} + I_{r,i,j} + C_{l,i,j} \frac{dV_{l,i+1,j}}{dt} + I_{l,i+1,j} - \sum_s \left(I_{b,i,j}^{(s)} + \sum_{\substack{k=1 \\ k \neq j}}^M I_{c,i,j,k}^{(s)} \right) = 0, \quad (\text{S2f})$$

$$I_{c,i,j,k}^{(s)} = g_{c,i,j,k} \left[\phi_{c,i,j} - \phi_{c,i,k} - \frac{RT}{z_s F} \ln \left(\frac{s_{c,i,k}}{s_{c,i,j}} \right) \right], \quad (\text{S2g})$$

$$I_{b,i,j}^{(s)} = g_{b,i,j} \left[\phi_{c,i,j} - \frac{RT}{z_s F} \ln \left(\frac{s_b}{s_{c,i,j}} \right) \right], \quad (\text{S2h})$$

$$\frac{ds_{c,i,j}}{dt} = \frac{1}{z_s F v_{i,j}} \left(I_{r,i,j}^{(s)} + I_{l,i+1,j}^{(s)} - I_{b,i,j}^{(s)} - \sum_{\substack{k=1 \\ k \neq j}}^M I_{c,i,j,k}^{(s)} \right). \quad (\text{S3})$$

We set

$$\eta_c^s = \phi_c + \frac{RT}{z_s F} \ln \frac{s_c^{(s)}}{s_*} \quad \text{and} \quad \eta_b^s = \frac{RT}{z_s F} \ln \frac{s_b^{(s)}}{s_*}.$$

Where s_* represents the unit concentration. $I_{c,i,j,k}^{(s)}$ and $I_{b,i,j}^{(s)}$ can be rewritten as

$$i_{c,i,j,k}^{(s)} = g_{c,i,j,k} (\eta_{c,j}^s - \eta_{c,k}^s), \quad i_{b,i,j}^{(s)} = g_{b,i,j} (\eta_{c,j}^s - \eta_b^s).$$

By applying the same method

$$I_c = \sum_{\substack{k=1 \\ k \neq j}}^M I_{c,i,j,k}^{(s)} = \int_{\partial w_j} \mathbf{J}_c \cdot \mathbf{n} \, d\ell.$$

Here

$$\mathbf{J}_c = -\sigma_c \nabla \eta_c^s.$$

We can write (S2f) as

$$c_{\text{pre}} \partial_t V_{\text{pre}} + i_{\text{pre}} + c_{\text{post}} \partial_t V_{\text{post}} + i_{\text{post}} - \sum_s \bar{g}_b (\eta_c^s - \eta_b^s) + \sum_s \nabla \Omega \cdot (\sigma_c \nabla \Omega \eta_c^s) = 0,$$

$$\partial_t s_c^s = \frac{1}{z_s F \delta_c} \left[i_{\text{pre}}^{(s)} + i_{\text{post}}^{(s)} - \bar{g}_b (\eta_c^s - \eta_b^s) + \nabla \Omega \cdot (\sigma_c \nabla \Omega \eta_c^s) \right].$$

We define

$$\alpha_s = \frac{RT}{z_s F}, \quad \eta_c^s = \phi_c + \alpha_s \ln \frac{s_c^{(s)}}{s_*}, \quad \eta_b^s = \phi_b + \alpha_s \ln \frac{s_b^{(s)}}{s_*}.$$

Therefore,

$$\eta_c^s - \eta_b^s = \phi_c - \phi_b + \alpha_s \ln \frac{s_c^{(s)}}{s_b^{(s)}},$$

and

$$\nabla \Omega \eta_c^s = \nabla \Omega \phi_c + \alpha_s \nabla \Omega \ln s_c^{(s)}.$$

Substituting these into the compact cleft equations gives

$$\begin{aligned} c_{\text{pre}} \partial_t V_{\text{pre}} + i_{\text{pre}} + c_{\text{post}} \partial_t V_{\text{post}} + i_{\text{post}} - \sum_s \bar{g}_b \left[\phi_c + \alpha_s \ln \frac{s_c^{(s)}}{s_b^{(s)}} \right] \\ + \sum_s \nabla \Omega \cdot \left[\sigma_c^{(s)} \left(\nabla \Omega \phi_c + \alpha_s \nabla \Omega \ln s_c^{(s)} \right) \right] = 0, \end{aligned} \quad (\text{S2f})$$

$$\begin{aligned} \partial_t s_c^{(s)} = \frac{1}{z_s F \delta_c} \left[i_{\text{pre}}^{(s)} + i_{\text{post}}^{(s)} - \bar{g}_b \left(\phi_c + \alpha_s \ln \frac{s_c^{(s)}}{s_b^{(s)}} \right) \right. \\ \left. + \nabla \Omega \cdot \left[\sigma_c^{(s)} \left(\nabla \Omega \phi_c + \alpha_s \nabla \Omega \ln s_c^{(s)} \right) \right] \right]. \end{aligned} \quad (\text{S3})$$

Here $\sigma_c^{(s)}$ is the cleft electro-diffusive conductivity of species s , kept as a coefficient.

Overall cleft form

Hence, we can have the overall cleft form

$$\left\{ \begin{array}{ll}
 0 = c_m \partial_t \phi_m + I_m + \sum_{N_{junctions}} [g_{myo}(\phi_m - \langle \phi_{pre} \rangle) + g_{myo}(\phi_m - \langle \phi_{post} \rangle)] & \text{Axial} \\
 0 = \bar{g}_{myo}(\phi_{pre} - \phi_m) + c_{pre} \partial_t V_{pre} + i_{pre} + \bar{g}_{gap}(\phi_{pre} - \phi_{post}) \\
 \quad - \nabla_{\Omega} \cdot (\sigma_{pre} \nabla_{\Omega} \phi_{pre}) & \text{Pre junctional} \\
 0 = \bar{g}_{myo}(\phi_{post} - \phi_m) + c_{post} \partial_t V_{post} + i_{post} + \bar{g}_{gap}(\phi_{post} - \phi_{pre}) \\
 \quad - \nabla_{\Omega} \cdot (\sigma_{post} \nabla_{\Omega} \phi_{post}) & \text{Post junctional} \\
 0 = c_{pre} \partial_t V_{pre} + i_{pre} + c_{post} \partial_t V_{post} + i_{post} - \sum_s \bar{g}_b \left[\phi_c + \alpha_s \ln \frac{s_c^{(s)}}{s_b^{(s)}} \right] \\
 \quad + \sum_s \nabla_{\Omega} \cdot \left[\sigma_c^{(s)} \left(\nabla_{\Omega} \phi_c + \alpha_s \nabla_{\Omega} \ln s_c^{(s)} \right) \right] & \text{Cleft} \\
 \partial_t s_c^{(s)} = \frac{1}{z_s F \delta_c} \left[i_{pre}^{(s)} + i_{post}^{(s)} - \bar{g}_b \left(\phi_c + \alpha_s \ln \frac{s_c^{(s)}}{s_b^{(s)}} \right) \right. \\
 \quad \left. + \nabla_{\Omega} \cdot \left[\sigma_c^{(s)} \left(\nabla_{\Omega} \phi_c + \alpha_s \nabla_{\Omega} \ln s_c^{(s)} \right) \right] \right] & \text{Cleft ionic concentration}
 \end{array} \right. \quad (1a)$$

DAE form of the ID–cleft continuum system

We first define the membrane voltages as

$$V_{pre} = \phi_{pre} - \phi_c, \quad V_{post} = \phi_{post} - \phi_c.$$

Therefore,

$$\phi_{pre} = V_{pre} + \phi_c, \quad \phi_{post} = V_{post} + \phi_c.$$

Let the state vector be

$$y = \begin{bmatrix} \phi_m \\ V_{pre} \\ V_{post} \\ \phi_c \\ \{s_c^{(s)}\}_{s \in \mathcal{S}} \end{bmatrix}, \quad \dot{y} = \begin{bmatrix} \partial_t \phi_m \\ \partial_t V_{pre} \\ \partial_t V_{post} \\ \partial_t \phi_c \\ \{\partial_t s_c^{(s)}\}_{s \in \mathcal{S}} \end{bmatrix}.$$

For compact notation, define

$$\mathcal{L}_{pre}(u) = \nabla_{\Omega} \cdot (\sigma_{pre} \nabla_{\Omega} u), \quad \mathcal{L}_{post}(u) = \nabla_{\Omega} \cdot (\sigma_{post} \nabla_{\Omega} u).$$

For each ionic species s , define the bulk exchange term and the lateral cleft transport term as

$$\mathcal{B}_s(\phi_c, s_c^{(s)}) = \bar{g}_b \left[\phi_c + \alpha_s \ln \left(\frac{s_c^{(s)}}{s_b^{(s)}} \right) \right],$$

and

$$\mathcal{D}_s(\phi_c, s_c^{(s)}) = \nabla_{\Omega} \cdot \left[\sigma_c^{(s)} \left(\nabla_{\Omega} \phi_c + \alpha_s \nabla_{\Omega} \ln s_c^{(s)} \right) \right].$$

The continuum ID–cleft system can then be written as a mass-matrix DAE:

$$M(y)\dot{y} = F(y),$$

where

$$M(y) = \begin{bmatrix} c_m & 0 & 0 & 0 & 0 \\ 0 & c_{\text{pre}} & 0 & 0 & 0 \\ 0 & 0 & c_{\text{post}} & 0 & 0 \\ 0 & c_{\text{pre}} & c_{\text{post}} & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{\mathcal{S}} \end{bmatrix}.$$

Here $I_{\mathcal{S}}$ is the identity operator acting on the ionic concentration variables $\{s_c^{(s)}\}_{s \in \mathcal{S}}$. The right-hand side is

$$F(y) = \begin{bmatrix} F_m(y) \\ F_{\text{pre}}(y) \\ F_{\text{post}}(y) \\ F_c(y) \\ \{F_s(y)\}_{s \in \mathcal{S}} \end{bmatrix},$$

with

$$\begin{aligned} F_m(y) &= -I_m - \sum_{\text{junctions}} [g_{\text{myo}}(\phi_m - \langle V_{\text{pre}} + \phi_c \rangle) + g_{\text{myo}}(\phi_m - \langle V_{\text{post}} + \phi_c \rangle)], \\ F_{\text{pre}}(y) &= -\bar{g}_{\text{myo}}(V_{\text{pre}} + \phi_c - \phi_m) - i_{\text{pre}} - \bar{g}_{\text{gap}}(V_{\text{pre}} - V_{\text{post}}) + \mathcal{L}_{\text{pre}}(V_{\text{pre}} + \phi_c), \\ F_{\text{post}}(y) &= -\bar{g}_{\text{myo}}(V_{\text{post}} + \phi_c - \phi_m) - i_{\text{post}} - \bar{g}_{\text{gap}}(V_{\text{post}} - V_{\text{pre}}) + \mathcal{L}_{\text{post}}(V_{\text{post}} + \phi_c), \\ F_c(y) &= -i_{\text{pre}} - i_{\text{post}} + \sum_{s \in \mathcal{S}} \mathcal{B}_s(\phi_c, s_c^{(s)}) - \sum_{s \in \mathcal{S}} \mathcal{D}_s(\phi_c, s_c^{(s)}), \end{aligned}$$

and

$$F_s(y) = \frac{1}{z_s F \delta_c} \left[i_{\text{pre}}^{(s)} + i_{\text{post}}^{(s)} - \mathcal{B}_s(\phi_c, s_c^{(s)}) + \mathcal{D}_s(\phi_c, s_c^{(s)}) \right].$$

Therefore, the complete mass-matrix form is

$$\begin{bmatrix} c_m & 0 & 0 & 0 & 0 \\ 0 & c_{\text{pre}} & 0 & 0 & 0 \\ 0 & 0 & c_{\text{post}} & 0 & 0 \\ 0 & c_{\text{pre}} & c_{\text{post}} & 0 & 0 \\ 0 & 0 & 0 & 0 & I_{\mathcal{S}} \end{bmatrix} \begin{bmatrix} \partial_t \phi_m \\ \partial_t V_{\text{pre}} \\ \partial_t V_{\text{post}} \\ \partial_t \phi_c \\ \{\partial_t s_c^{(s)}\}_{s \in \mathcal{S}} \end{bmatrix} = \begin{bmatrix} F_m(y) \\ F_{\text{pre}}(y) \\ F_{\text{post}}(y) \\ F_c(y) \\ \{F_s(y)\}_{s \in \mathcal{S}} \end{bmatrix}$$

This form shows that the ID–cleft continuum system is a DAE rather than a traditional ODE system. In particular, the cleft potential ϕ_c has no independent evolution equation of the form $\partial_t \phi_c = f(y)$. Instead, it is determined by the charge-conservation constraint. Therefore, the mass matrix is singular.

The general DAE form is

$$\mathcal{G}(t, y, \dot{y}) = 0.$$

For a linearly implicit system, this can be written as

$$\mathcal{G}(t, y, \dot{y}) = M(t, y)\dot{y} - F(t, y) = 0,$$

where $M(t, y) = \partial \mathcal{G} / \partial \dot{y}$ is the mass matrix. If $M(t, y)$ is nonsingular, the system can be reduced to an explicit ODE. In contrast, if $M(t, y)$ is singular, some components of y are not governed by independent evolution equations and are instead determined by algebraic constraints. Therefore, the system is a DAE.

In the ID-cleft model, the cleft potential ϕ_c is determined by the Kirchhoff-type current conservation constraint rather than by an independent equation for $\partial_t \phi_c$. Hence, the column of the mass matrix associated with $\partial_t \phi_c$ is zero, which makes $M(t, y)$ singular.